

RIPHAH INTERNATIONAL UNIVERSITY



Faculty of Computing FINAL YEAR PROJECT INITIAL PROPOSAL

CareLink

Project Team

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Project Proposal

Project Title: CareLink: Agentic AI-Powered Medication Adherence & Chronic-Care Coordination System

Description:

Problem Statement

Medication non-adherence is a common issue, particularly among elderly patients and people managing long term treatments. Patients may intentionally or accidentally miss a dose, take the medicine incorrectly, or disregard the set schedule. Moreover, caregivers and family members are often not aware of such deviations. At the same time, nurses and coordinators responsible for the patient's supervision cannot monitor the adherence constantly and in real-time. The medication reminder apps currently available on the market are primarily designed to notify patients of the time for dosing. However, such notifications do not evaluate the adherence status and level of care needed, and they do not assess whether the patient independently manages their schedule or requires assistance. Therefore, there is a need for a technology which monitors medication administration, recognizes patterns of non-adherence, and appropriately responds by suggesting particular interventions or informing a caregiver/nurse.

Proposed Solution

CareLink is a web and app based healthcare platform utilizing agentic AI to help with medication adherence and general care coordination. The system would oversee a patient's medication schedule and adherence records while also having AI agents that watch out for signs of any deviations and address them in context-appropriate ways. The same medication adherence issues will not be addressed in the same way if the patient acts differently around the time of a missed dose. The AI will be able to make a call on if another reminder is needed, if a caregiver has to be alerted, or if the case has to be escalated to a nurse or care coordinator. A controlled AI assistant can be used to answer any questions a patient might have about their medication and care. The most critical decisions will be audited to ensure they are not prone to any bias and that the AI adheres to safety protocols. The system will be designed to be human-centric and will be focused on empowering patients, caregivers, and care coordinators rather than replacing them.

Key Features

Core features would include patient and medication management with medication names, doses, schedules, and adherence data. Core features also include an AI-aided analysis of medication-taking patterns to determine potential risks of poor adherence. The system would offer adaptive interventions, which could include dynamically adjusted reminders and follow-up schedules based on a patient's response and adherence patterns. An agentic care-coordination workflow would be able to prioritize cases requiring the intervention of human caregivers by escalating them with a notification to the relevant caregivers or care coordination personnel. The system would aim to provide human-level explainability for risk assessment and intervention decisions to explain the logic behind alerts and escalations. A prescription/document processing component would be useful in extracting medication orders from scanned or uploaded documents, with the possibility to have them reviewed and approved by humans prior to being entered into medication schedules within the system. Where possible and where a relevant wearable API is available, other vital signs, such as heart rate, could be incorporated as additional inputs for threshold-based caregiver notifications.

Technical Approach

The system will use a modern web and app based architecture with a frontend for patients, caregivers and care coordinators, and a backend responsible for secure data and workflow management. The backend will provide APIs for patient records, medications, adherence events, alerts and care coordination. A relational database such as PostgreSQL will store critical patients and medication data. The AI layer will use multiple specialized agents, including an Adherence Agent, Intervention Agent, and Care Coordinator Agent. These agents will work with controlled tools and predefined safety rules rather than directly changing critical records. Machine learning and AI techniques will be used for adherence to pattern analysis and risk prediction. The system will also maintain explanations and logs for important AI decisions. A notification layer will handle reminders and caregiver/care-coordinator alerts. If a wearable integration is feasible, its available API will be connected through a controlled backend service. The system will follow a human-in-the-loop design, meaning that AI recommendations can be reviewed or overridden by authorized users. Critical healthcare decisions will not be left entirely to the AI system.

Objectives & Expected Outcomes

The project aims to provide a centralized system for managing medication schedules and monitoring medication adherence. The project also aims to identify changing adherence behavior earlier than a simple reminder system and provide appropriate interventions based on the patient's situation. The system aims to reduce unnecessary caregiver or nurse alerts by escalating cases according to risk and predefined safety conditions. The project also aims to demonstrate how agentic AI can coordinate multiple steps of a healthcare support workflow while maintaining

explainability, safety controls, and human oversight. The platform will be delivered as an MVP over the two-semester development cycle. The implementation will focus primarily on the core agentic adherence and care-coordination workflow, while advanced features such as extensive wearable integrations, advanced voice support and other healthcare integrations will remain optional or future work.